

FINDING SLOPE USING GRAPHS AND TABLES



LESSON INTRODUCTION

MA.8.AR.3.2 Given a table, graph, or written description of a linear relationship, determine the slope.

LEARNING TARGETS

- I can determine the slope of a line from a graph or a table, given a mathematical context.
- I can relate the slope as the rate of change of the relationship between variables.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

WORKING TOWARD

MA.912.AR.2.2 Write a linear two-variable equation to represent the relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function and determine and interpret its key features.

ESSENTIAL QUESTIONS

- How do you determine the slope of a line from a graph?
- How do you determine the slope of a line from a table?
- Is there only one way to determine the slope of a line? Explain your answer.
- What is the rate of change? Share an example of a rate of change.

LESSON NARRATIVE

Students will revisit the concept of slope, $\frac{\text{change in } y}{\text{change in } x}$, and develop a deeper understanding of the concept. Students will find the slope from a line from a graph and from a table of values. Students will also relate the slope of a line to the rate of change in a linear relationship.

COMPONENT SUMMARY

7.1.1 WARM-UP (~5 MIN.)

Justify which rate is preferred for reading a book and why

7.1.3 COLLABORATION (10 MIN.)

Collaboratively analyze students' work calculating slope to uncover that which point is used first does not matter

7.1.5 YOUR TURN! (5-10 MIN.)

Determine the slopes of lines from graphs and tables of values

7.1.2 REFRESHER (10 MIN.)

Use the ratio of vertical lengths to horizontal lengths in similar triangles to determine the slope of a line

7.1.4 GUIDED INSTRUCTION (15 MIN.)

Connect slope to the rate of change and relate the sign of the slope with the direction of the rate of change

7.1.6 WRAP-UP (~5 MIN.)

Reflect on level of understanding of the learning targets

RESOURCES

GLOSSARY

Key Terms:

- Slope
- Rate of change

Additional Terms: N/A

MATERIALS

- (Optional) Notecards for card sort
- (Optional) Graph paper

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly calculate slope as $\frac{\text{change in } x}{\text{change in } y}$.
- Students may struggle with staying consistent with their ordered pairs when finding $\frac{\text{change in } y}{\text{change in } x}$.
- Students may incorrectly identify the ordered pairs of points on lines in graphs.

THINGS TO CONSIDER

- Consider allowing students to use calculators to calculate the slope if they struggle with integer operations. Having students write and read the calculations will reinforce facts and help build their skills.
- Consider having students describe whether the slope is positive, negative, or zero prior to performing the calculation as a way to check for the reasonableness of their computation.
- Avoid discussing the "slope formula." Students struggle with the notation (x_1, x_2, y_1, y_1) . Instead, focus on the concept of slope being the rate of change in y over the change in x .
- Undefined slopes are further developed in Algebra 1 and do not need to be emphasized in 8th grade.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Consider implementing a Think-Pair-Share routine during 7.1.3 Collaboration. Students will have the opportunity to discuss whether the order of the points matters when calculating the slope.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.K12.MTR.6.1 Reasonableness**

During 7.1.4 Guided Instruction, have students discuss how they can tell from a graph or table whether the slope will be positive, negative, or zero. They can then use this understanding to assess the reasonableness of the slopes they calculate in other questions.

DIFFERENTIATE INSTRUCTION

During 7.1.5 Your Turn!, students are asked to complete two questions calculating the slope from graphs and two questions calculating the slope from tables. Consider allowing students to choose which one of the two questions they wish to complete for each type. Since these questions are computational, consider asking advanced students to write a letter to a friend explaining how they calculated the slope. For struggling students, consider making this activity into a card sort with all the answers already filled in so they can focus on matching the answers.

Drawing slope triangles in 7.1.4 Guided Instruction will provide a visual entry point. Discussions in 7.1.3 Collaboration provide an auditory entry point.

7.1.1 WARM-UP: WOULD YOU RATHER?**TEACHER MOVES**

Ask students to share their reasonings for the options they choose. Help students see the relevance of the use of ratios to determine and support their decisions.

ENRICHMENT

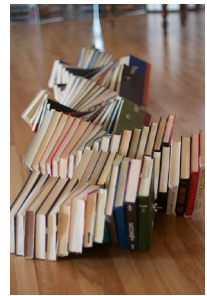
Consider asking students the following:

- What mathematical operations or concepts did you use to determine your answer?
- Is there a way to represent the answer you chose on a graph?

**7.1.1 Warm-Up: Would You Rather?**

1. Would you rather read 12 pages each night in a book with 144 total pages or read 50 pages three times a week in a book with 132 total pages?

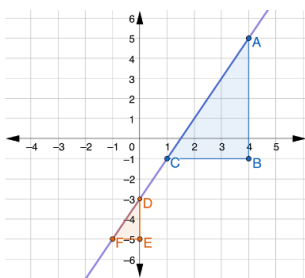
Explain your reasoning using mathematics for whichever option you choose.



Answers will vary. Arguments could be made for either, but should be mathematically based.

7.1.2 Refresher: Using Similar Triangles to Find the Slope of a Line

Similar triangles can be used to determine the slope of a line by comparing the ratio of vertical lengths to horizontal lengths of the triangles.



- For each triangle, label the ordered pairs of each vertex.
Blue Triangle: A (4, 5), B (4, -1), C (1, -1)
Orange Triangle: D (0, -3), E (0, -5), F (-1, -5)
- What is the length of the vertical leg of the blue triangle?
6 units
- How could you use the coordinates of the vertices of the vertical leg to find its length?
Subtract the y-values.
 $5 - (-1) = 6$ or $-1 - 5 = |-6| = 6$
- What is the length of the horizontal leg of the blue triangle?
3 units
- How could you use the coordinates of the vertices of the horizontal leg to find its length?
Subtract the x-values.
 $4 - 1 = 3$ or $1 - 4 = |-3| = 3$



The **slope** of a line is the ratio of the change in the vertical direction to change in the horizontal direction, often expressed as the $\frac{\text{change in } y}{\text{change in } x}$, or $\frac{\Delta y}{\Delta x}$.

- Complete the statement.
In the equation $\text{slope} = \frac{\text{change in } y}{\text{change in } x}$, the ○ change in y
○ change in x
relates to finding the length of the vertical leg of the triangle and the ○ change in y
○ change in x relates to finding the length of the horizontal leg of the triangle.
- Use the coordinates of the vertices of the blue triangle to find the slope of the purple line. Show your work or explain what you did.
 $\text{Slope} = \frac{5 - (-1)}{4 - 1} = \frac{6}{3} = 2$
- Use the coordinates of the vertices of the orange triangle to find the slope of the purple line. Show your work or explain what you did.
 $\text{Slope} = \frac{-3 - (-5)}{0 - (-1)} = \frac{2}{1} = 2$
- Check your answers with your partner.
Answers will vary. Encourage students to discuss why the slope is the same for both triangles.
- Make a conjecture about whether the order in which you subtract to find the change in x and change in y matters.
Answers will vary. Order does not matter but will be addressed in the next section so it is not imperative that it is resolved here.

7.1.2 REFRESHER: USING SIMILAR TRIANGLES TO FIND THE SLOPE OF A LINE

QUESTIONING

Consider having self-directed learners complete all of the questions independently before debriefing with the whole class. During the discussion, check understanding that any two points on the line can be used to determine the slope.

TEACHER MOVES

Ask students to show their work and explain their reasoning. question 10 asks whether the order in which we subtract matters when finding the change in x and y. This topic is addressed thoroughly in 7.1.3 Collaboration.

ENRICHMENT

Consider asking students the following:

- Would the slopes of two triangles on different lines be the same as well?
- What does this slope tell us about the line?

7.1.3 COLLABORATION: FINDING SLOPE FROM GRAPHS

TEACHER MOVES

Ask students to share their thinking for questions 2 and 3 with a partner. Be sure students understand that although it doesn't matter which point they start with, order does need to be consistent for calculating both the change in y and change in x. If possible, call on a student to share these takeaways with the whole group (rather than the teacher) to develop student math identity.

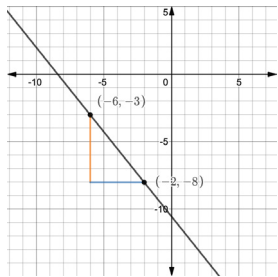
DIFFERENTIATE INSTRUCTION

For students who are struggling, remember the goal is to help students understand the concept of slope. Consider providing a calculator to help the students focus on the concept of slope rather than computation. Additional practice calculating the slope is provided in 7.1.5 Your Turn!



7.1.3 Collaboration: Finding Slope from Graphs

Fatima and Oliver both drew the same triangle using the points $(-6, -3)$ and $(-2, -8)$ on the line to determine the slope, but they performed the calculations differently. Their partial work is shown.



- With your partner, complete each student's work and answer the questions that follow.

Fatima's Work	Oliver's Work
$\text{slope} = \frac{\text{change in } y}{\text{change in } x}$	$\text{slope} = \frac{\text{change in } y}{\text{change in } x}$
$\text{slope} = \frac{-3 - (-8)}{-6 - (-2)}$	$\text{slope} = \frac{-8 - (-3)}{-2 - (-6)}$
$\text{slope} = \frac{5}{-4}$	$\text{slope} = \frac{5}{-4}$

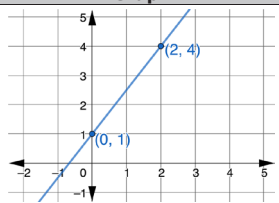
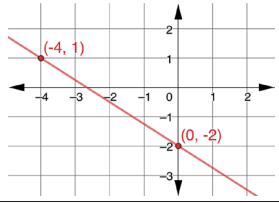
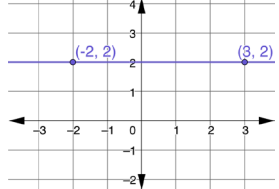
- Fatima compared her work to Oliver's and said, "It doesn't matter which order we subtracted the values. We found the same slope." Explain why you agree or disagree.
Answers will vary. Fatima and Oliver started with different points and found the same slope.
- Oliver said, "It doesn't matter which order you subtract, but whichever coordinate pair you start with to find the change in y, you have to also start with its x-value when finding the change in x." What do you think Oliver meant?
Answers will vary. Whichever point you start with when calculating the change in y, you need to start with the x-value of that point with calculating the change in x.



7.1.4 Guided Instruction: Finding Slope from Graphs and Tables

The slope of any line can be determined from a graph using any two points on the line.

Three graphs and the calculations used to determine their slope are shown.

Graph	Calculation	Slope
	$m = \frac{4 - 1}{2 - 0} = \frac{3}{2}$	$\frac{3}{2}$
	$m = \frac{1 - (-2)}{-4 - 0} = \frac{3}{-4}$	$-\frac{3}{4}$
	$m = \frac{2 - 2}{3 - (-2)} = \frac{0}{5}$	0

7.1.4 GUIDED INSTRUCTION: FINDING SLOPE FROM GRAPHS AND TABLES

DIFFERENTIATE INSTRUCTION

Drawing slope triangles in 7.1.4 Guided Instruction will provide a visual entry point and connect to previous components. Discussions will also provide an auditory entry point.

1. Use the graphs and their slopes to complete the summary.
- The slope of a line can be positive, negative, or 0.
- The slope is positive when the line
- does not rise or fall
 - is increasing, or rising,
 - is decreasing, or falling,

from left to right. The
- slope is negative when the line
- does not rise or fall
 - is increasing, or rising,
 - is decreasing, or falling,

from left to right. The
- slope is 0 when the line
- does not rise or fall
 - is increasing, or rising,
 - is decreasing, or falling,

from left to right.

Recall that linear relationships can also be represented using tables of values. The table to the right represents a linear relationship.

x	y
-4	-9
-2	-4
0	1
2	6
4	11

Notice that the x-values in the table are increasing by the same amount in each row.

2. How are the y-values in the table changing as the x-values increase by 2?

The y-values are increasing by 5 as the x-values increase by 2.

ENRICHMENT

Consider asking students the following:

- How can you tell from looking at the graph of a line whether the slope will be positive, negative, or zero?
- Can a vertical line have a slope? Explain why not.

7.1.4 GUIDED INSTRUCTION: FINDING SLOPE FROM GRAPHS AND TABLES (CONTINUED)

QUESTIONING

Thinking about the concept of rate of change, ask students to come up with examples and non-examples of constant rates of change (in addition to what is provided in the lesson). Ask students what the graph might look like if the rate is not constant for a non-example they provide.

3. With your partner, decide who will be Partner A and who will be Partner B. Then, find the slope of the linear relationship using the ordered pairs indicated below.
 - a. Show your work in the space provided.

Partner A	Partner B
Find the slope using the ordered pairs $(-2, -4)$ and $(0, 1)$ from the table. $\text{Slope} = \frac{-4 - 1}{-2 - 0} = \frac{-5}{-2} = \frac{5}{2}$	Find the slope using the ordered pairs $(4, 11)$ and $(2, 6)$ from the table. $\text{Slope} = \frac{11 - 6}{4 - 2} = \frac{5}{2}$

- b. Check your work with your partner.

The slope of any line can be determined from a table of values using any two ordered pairs in the linear relationship because the **rate of change** in a linear relationship is constant.



The **rate of change** is the ratio of change in one quantity to the corresponding change in another quantity.

A rate, such as 45 miles per hour, gives the speed of an object. The rate of change of a line tells us how fast or how slow a line is increasing or decreasing. Rates can be constant. For example, the speed of a horse on a carousel is constant while the ride is in motion, but the speed of someone riding on a merry-go-round is not constant because it can change depending on how fast or slow the merry-go-round is being pushed.



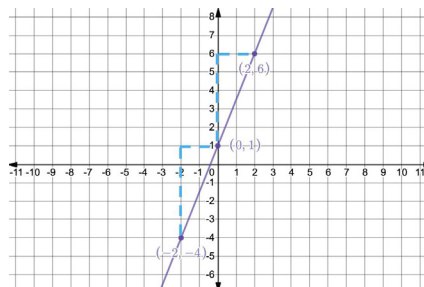
TEACHER MOVES

Question 4 consolidates the key topics addressed in this component. Check for understanding by having a variety of students share. Involve more students by asking them to signal if they agree or disagree with statements shared. Promote active listening by calling on students and asking them to explain what another student shared in their own words.

4. Complete the summary to describe the relationships between the quantities observed for the linear relationship represented by the table of values.

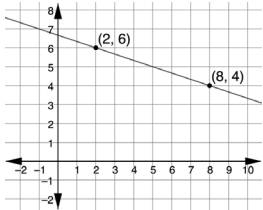
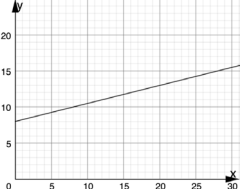
The y -values in the table ☐ increase by 2 ☒ increase by 5 each time
 the x -values ☐ increase by 2 ☐ increase by 5. Slope is the
☐ change in x over the ☐ change in x , or
☒ change in y ☐ change in y
 in this case. The slope, or rate of change, is constant
 for linear relationships, so the slope is
☒ the same, regardless of ☐ different, depending on which two ordered pairs

on the line are used to calculate it. This means that the line is increasing or decreasing at the same rate between every point along the line. That is why it forms a straight line, as shown on the graph.



7.1.5 Your Turn!

1. Determine the slope of each linear relationship.

Linear Relationship	Calculation	Slope										
	<i>Answers will vary.</i> $\frac{6 - 4}{2 - 8} = \frac{2}{-6}$	$-\frac{1}{3}$										
<table border="1" data-bbox="250 533 412 648"><thead><tr><th>x</th><th>y</th></tr></thead><tbody><tr><td>-2</td><td>-11</td></tr><tr><td>-1</td><td>-8</td></tr><tr><td>0</td><td>-5</td></tr><tr><td>1</td><td>-2</td></tr></tbody></table>	x	y	-2	-11	-1	-8	0	-5	1	-2	<i>Answers will vary.</i> $\frac{-11 - (-8)}{-2 - (-1)} = \frac{-3}{-1}$	$\frac{3}{1}$
x	y											
-2	-11											
-1	-8											
0	-5											
1	-2											
	<i>Answers will vary.</i> $\frac{8 - 10}{0 - 8} = \frac{-2}{-8}$	$\frac{1}{4}$										
<table border="1" data-bbox="250 890 412 995"><thead><tr><th>x</th><th>y</th></tr></thead><tbody><tr><td>-3</td><td>-5</td></tr><tr><td>0</td><td>-5</td></tr><tr><td>3</td><td>-5</td></tr><tr><td>6</td><td>-5</td></tr></tbody></table>	x	y	-3	-5	0	-5	3	-5	6	-5	<i>Answers will vary.</i> $\frac{-5 - (-5)}{-3 - 0} = \frac{0}{-3}$	0
x	y											
-3	-5											
0	-5											
3	-5											
6	-5											

7.1.5 YOUR TURN!

DIFFERENTIATE INSTRUCTION

For students who are struggling, consider making this activity into a card sort with all the answers already filled in so they can focus on matching the answers.

Also, consider allowing students to choose which one of the two questions they wish to complete for each type.

QUESTIONING

To push students further with the concept of slope, add a blank row at the bottom of the table and challenge the students to create and work out their own questions using either a graph or table of values. The questions are all computational and review concepts addressed in the previous unit. Consider asking students who have mastered this skill to write a letter to a friend explaining how they calculated the slope.

7.1.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current level of understanding is related to each target.
Answers will vary.

1. I can determine the slope of a line from a graph or a table, given a mathematical context.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.
2. I can relate the slope to the rate of change of the relationship between variables.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.

7.1.6 WRAP-UP: REFLECTION

QUESTIONING

Consider creating a chart based on student responses. Put an x on which level of understanding students chose. Then, analyze the responses to see where most of student understanding lies. If most students understand the content, they may be equipped to move on. However, if most students check that they need help, it may be best to review the lesson before moving on to solidify the concepts.

"I can determine the slope of a line from a graph or a table, given a mathematical context."	I need help.	I'm getting there.	I understand this.
Student 1			
Student 2			
Student 3			